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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 15: EIA Process in India

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Environment (Protection) Act, 1986

This act is the legal base for EIA process in India.

It states that
"The Central Government, shall have the power to take all such measures as it deems necessary or expedient for the purpose of protecting and improving the quality of the environment and preventing controlling and abating environmental pollution."

National Forest Policy, 1988

- Launched with objective of ensuring environmental stability and ecological balance essential for the sustenance of all life forms – human, animal & plants.
- Targets to bring one third area under tree/plantation cover.

The protection of environment has been provided in Articles 48 A and 51 A (g) of Indian constitution. Time to time strengthened by judicial interpretation of Article 21 (Protection of life & personal liberty).



National Conservation Strategy & Policy, 1992

- Sustainable and equitable use of resources
- Conserve natural and man-made heritage
- Prevent & control future deterioration in land, water and air
- Development projects: minimize their adverse environmental consequences
- Restoration of ecologically degraded areas

Policy Statement on Abatement of Pollution,1992

- Emphasis on pollution prevention in place of the conventional end-of-the-pipe treatment
- Adoption of best available and practicable technologies for pollution prevention
- Promotion of clean and low waste technologies, waste minimization, reuse or recycling, improvement of water quality, environment audit, natural resource accounting etc.



National Water Policy, 2012

The objective of the National Water Policy

- Take cognizance of the existing situation
- Propose a framework for creation of a system of laws and institutions
- Plan of action with unified national perspective



National Environmental Policy (NEP)-2006

- Better livelihoods through resources conservation instead of their degradation
- Greater role of and responsibilities to local bodies
- EIA act as principal methodology for appraisal and review of new projects
- Coordination among stakeholders for environmental management
- Environmental protection as an integral part of the development

EIA Notification2006

- Accordance with the objectives of National Environment Policy
- Legally based on Environment (Protection) Act, 1986
- Puts some certain restrictions and prohibitions on new projects; modernization of existing projects or activities
- Applicable in every part of India (including territorial waters)
- Restrictions based on potential environmental impacts



State Environment Impact Assessment Authority (SEIAC)

- Constituted by the Central Government
- Three Members including a Chairman & Member – Secretary
- State/UT shall forward the names of the Members and member-secretary
- Members shall be professionals or experts
- One of the member shall be the Chairman
- Decisions of the committee shall be unanimous and taken in a meeting otherwise majority prevails
- Constituted after every 3 years

Need of Environment Clearance (EC)

- Environment clearance is a legal requirement to the project proponent which permits to start the project.
- EC needed before any construction work, or preparation of land by the project management except for securing the land.

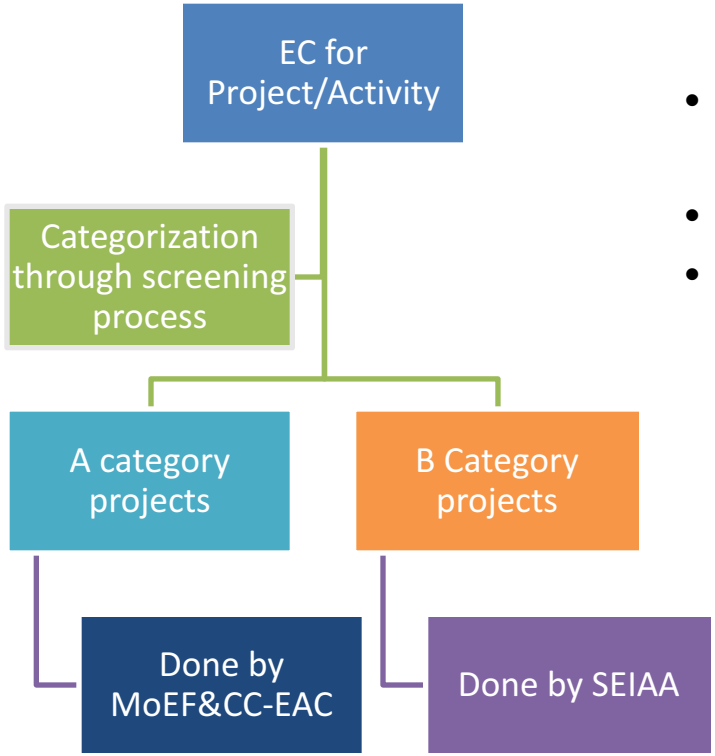
Projects requiring Environment Clearance (EC)

Environment Clearance needed for-

- All new projects or activities listed in the EIA notification 2006
- Expansion in activities listed in the EIA notification 2006 with addition of capacity beyond the limits specified
- If any change in product - mix in an existing manufacturing unit beyond the specified range



Process for Environment Clearance (EC)



- MoEF&CC: Ministry of Environment, Forest & Climate Change
- EAC: Environment Assessment Committee
- SEIAA: State Environment Impact Assessment Authority

What are the project categories?

- Projects are divided in categories depending upon their possible level of impacts (High impact projects, medium impact project etc.)
- For administrative ease, a guideline is created to clearly categorize the projects
- In EIA notification 2006, projects are divided in two categories (A & B)
  - Category A: High impact projects
  - Category B: Medium impact projects

Categorization of transport sector project in EIA notification 2006

Project Type	Category A	Category B
Airport	All airports	-
Oil & gas (Pipeline) transport	All oil & gas transport projects	-
Aerial ropeway		All aerial ropeway project
Harbor/Port	>5 million TPA of cargo handling capacity (excluding fishing harbors)	<5 million TPA of cargo handling capacity

TPA: Tonnes per annum

Categorization of highway project in EIA notification 2006

Project Type	Category A	Category B
Highway	New national highways and expansion of National highways greater than 30 KM, involving additional right of way greater than 20m involving land acquisition and passing through more than one State	New state highways and expansion of National / State highways greater than 30 km involving additional right of way greater than 20m involving land acquisition

Railway projects don't come under EIA notification 2006 but EIA report is necessary for other clearances such as forest clearance etc.

Change in project category in EIA notification 2006

Any project or activity specified in Category 'B' will be treated as Category A, if located in whole or in part within 10 km from the boundary of :

- Protected Areas notified under the Wildlife (Protection) Act, 1972
- Notified Eco-sensitive areas
- Critically Polluted areas as notified by the Central Pollution Control Board
- Inter-State boundaries and international boundaries

Application for Environmental Clearance (EC)

- Application seeking prior environmental clearance in all cases shall be made in the prescribed Form 1/Form 1A, if applicable.
- The applicant shall furnish, along with the application, a copy of the pre-feasibility project report.
- Pre-feasibility project report won't be required in case of few projects, then a copy of the conceptual plan shall be provided by applicant.



Form 1 & 1A in application for environment clearance

- Form 1A is a check-list form required from the applicant.
- Yes/No check-list about impact with details required in case of yes.
- 93 parameters in 10 categories of impact.
- Form 1A requires to fill details about 10 impacts such as vegetation, air quality, water quality.

Pre-feasibility report

It consists:

- Project background
- Project Description
- Site Analysis
- Proposed Infrastructure
- Rehabilitation & Resettlement Plan
- Project schedule & cost estimates
- Overall analysis of proposal (in terms of social, economic & environmental impacts)



Conceptual plan

It consists:

- Project description
- Site location
- Important features of project
- Impacts (if any)



Steps for Scoping Process in EIA notification 2006 1/2

- Comprehensive Terms Of Reference (ToR) for the preparation of an EIA report by applicant
- Site visit by a sub- group of appraisal committee
- ToR shall be conveyed by the appraisal committee within sixty days of the receipt of application

Scoping Process in EIA notification 2006 2/2

- If ToR (Terms of Reference) are not finalized and conveyed to the applicant within sixty days, provided ToR will be assumed final
- Approved ToR shall be displayed on the website of the MoEF&CC (Ministry of Environment, Forest & Climate Change)
- Prior environment clearance may be rejected on the recommendation of the appraisal committee
- Reason of rejection shall be communicated within 60 days



Public Consultations

- Public consultation through on site (public hearing) & obtain response in writing from concerned persons
- Public hearing conducted by SPCB (State Pollution Control board) or UTPCC (Union Territory Pollution Control Committee)
- In absence of SPCB or UTPCC, an independent agency can be assigned by EAC/SEAC for public consultation.
- In case public hearing not possible due to any reason, appraisal committee shall mention in the report and start further appraisal

Importance of public consultation : Bhatta-Parsaul Case

In year 2011, U.P. govt. done acquisition of land for Yamuna expressway around Greater Noida (U.P.)

This caused huge uproar in surrounding region of project site and violence broke out.

A political & social crisis was created.



Image Source: amarujala.com

Do you think public participation can ease out project implementation and reduces risks for project ? (Yes, it does)

Exception cases for Public Consultations in EIA notification 2006

All projects or activities mentioned in EIA notification 2006 shall undertake Public Consultation except:

- Expansion of Roads and Highways which do not involve any further acquisition of land
- Projects or activities concerning national defense and security, or involving other strategic considerations as determined by the Central Government
- Specified project within existing industrial estates & Irrigation project



Steps for appraisal process in EIA notification 2006

- Grant environmental clearance on stipulated terms and conditions, or rejection of the application for environmental clearance, together with reasons for the same
- Based on the prescribed application Form 1 and Form 1A and addition information required after site visit by appraisal committee
- Recommendations of the appraisal committee shall be placed before the Ministry of Environment, Forest & Climate Change (MoEF&CC) for a final decision within the next fifteen days
- The appraisal of an application be shall be completed within sixty days of the receipt of the final EIA report



Post Environmental Clearance Monitoring

- Need to submit half-yearly compliance reports in respect of the stipulated prior environmental clearance terms and conditions in hard and soft copies, on 1st June and 1st December of each calendar year.

- All such compliance reports submitted by the project management shall be public documents.

Transferability of Environmental Clearance

- A prior environmental clearance granted may be transferred during its validity to another legal person entitled to undertake the project or activity on application with a written “no objection” by the transferor.

- It on the same terms and conditions under which the prior environmental clearance was initially granted, and for the same validity period.

- No reference to the Expert Appraisal Committee or State Level Expert Appraisal Committee concerned is necessary in such cases.

EIA Notification 2020 Draft

- Central govt. notified a new draft for EIA process in March 2020 for public comments/feedback.
- Notification shall come into force on the date of publication of final notification after changes (if any) in the Official Gazette.
- It is also legally based on Environment (Protection) act, 1986.



Major Differences between EIA notification 2006 & EIA draft 2020

EIA notification 2006	EIA draft notification 2020
• Only 2 project categories (A/B)	• 3 project categories (A/B1/B2)
• Various terms not defined	• Terms are defined
• Yearly data is required of impacts	• Data of one season is sufficient
• Monitoring at every 6 month	• Yearly monitoring
• Monitoring report by third party	• Monitoring report submitted by project proponent

Increment in project categories in EIA draft 2020

- In EIA notification 2006, few project under category B were required to be appraised at central level (or treated as category A projects).

- To ease out this, now in EIA draft 2020, category B projects are further classified in B1 & B2 categories.



Categorization of transport sector projects in EIA draft notification 2020

Project Type	A	B1	B2
Airports and Heliports including terrestrial and water ports	All airports	-	Water - aerodromes which are for commercial use Heliports which are for commercial use.
Oil & gas (Pipeline) transport	All oil & gas transport projects	-	-
Elevated roads or standalone flyovers or bridges	-	-	>1,50,000 sq. meters. of built-up area

Categorization of transport sector projects in EIA draft notification 2020

Project Type	A	B1	B2
Aerial ropeway			All projects located in notified ecologically sensitive areas.
Harbor/Port	Greater then 5 million TPA (Tonnes per annum) of cargo handling capacity (excluding fishing harbors)	Less then5 million TPA of cargo handling capacity Grater then 30000 ton per annum of fish handling capacity.	All projects in respect of Inland water ways



Categorization of highway projects in EIA draft notification 2020

Project	A	B1	B2
Highway/ Expressways or Multi-modal corridors or Ring Roads	New National Highways or Expressways or Multi-modal corridors or Ring Roads. Expansion or widening of existing National Highways or Expressways or Multi-modal corridors or Ring Roads by length more than 100 km involving widening or right of way more than 70 m on existing alignments or re-alignments or by-passes.	All new State Highway projects State Highway expansion projects in hilly terrain (above 1,000 meter above mean sea level).	Expansion or widening of existing National Highways or Expressways or Multi-modal corridors or Ring Roads by length between 25 km and 100 km involving widening or right of way more than 70 m on existing alignments or realignments or bypasses. Expansion or widening of existing State Highways (500 m to 1000 m above mean sea level)

Definition in EIA draft 2020: Accredited Environment Impact Assessment Consultant Organization (ACO)

Organization that is accredited with the

- National Accreditation Board for Education and Training (NABET) of Quality Council of India (QCI) or
- Any other agency, as may be notified by the Ministry from time to time

Was not defined in EIA notification 2006

Definition in EIA draft 2020 : Capital Dredging & Border Area

Capital Dredging

- Done during waterways/harbor project construction.
- one time process involving removal of virgin material from the sea-bed to create or deepen a shipping channel in order to serve larger ships. This includes dredging activity inside and outside the ports or harbors and Channels.

Was not defined in EIA notification 2006

Border Area

- Area falling within 100 kilometers aerial distance from the Line of Actual Control with bordering countries of India.

Definitions in EIA draft 2020 : Certificate of green buildings

Certificate issued for green building under the rating programme of:

- Green Rating for Integrated Habitat Assessment (GRIHA)
- Indian Green Building Council (IGBC)
- Leadership in Energy and Environmental Design (LEED)
- Excellence in Design for Greater Efficiencies (EDGE)

Was not defined in EIA notification 2006

Definitions: District/Divisional Level Expert Appraisal Committee

District/Divisional Level Expert Appraisal Committee

Committee of experts constituted at district level or at divisional level on the recommendation of the State Government or Union Territory administration, for environment appraisal of projects referred to it and for making appropriate recommendations.

Was not defined in EIA notification 2006

Exceptions for assessment of B1 category projects for EIA draft 2020

Any project or activity specified in Category 'B1' shall be appraised at the Central Level, if located in whole or in part,

Within 5/10 km (varies with type of project) from the boundary of-

- Protected Areas
- Critically Polluted Area
- Eco-sensitive area
- Inter State or Union Territory

Within the boundary of-

- Severely Polluted Area
- Eco-sensitive Zone

EIA draft 2020 : Reduced Monitoring compliance & Monitoring mechanism

Authorities were earlier mandated to monitor projects for compliance with environmental norms every six months. Now this has been changed to once in a year.

The monitoring mechanism is changed: Compliance report will be filed by the project proponents. This may give a chance for discrepancies.



Example of weak monitoring process: Vizag Gas Leak

- Polymer Plant in Vishakhapatnam
- Styrene gas leak happened on May 7.
- Plant running for over two decades without environmental & safety clearances.
- 11 people died and approx. 1000 people got sick.



Image Source: Indianexpress.com

Example of weak monitoring process: Refinery in Assam

- Natural gas in eastern Assam's Tinsukia district had a blowout and caught fire on May 2020
- Severe damage to the livelihoods & biodiversity.
- Operating for over 15 years without obtaining prior environmental & safety clearance.



Image Source: Indianexpress.com

EIA draft 2020: Change in requirement of Seasonal Data

New draft EIA notification 2020 permits to collect data for any environmental impact during one season (except for river valley projects).

Environmental data (parameters related to air quality, water quality etc.) changes according to seasons. It may hamper the baseline data and thus assessment and mitigation.



Example of change in impact assessment due to seasonal data: Air Pollution



- In summer season, due to low pressure, there is a chance that, pollutants will be in lesser compared to another season.
- It may impact baseline data.

Image Source: climatekids.nasa.gov

Change in impact assessment due to seasonal data: Water Pollution

- In summer season, due to significance change in flow in rivers (mostly in Himalayan river), may hamper the baseline data.



Conclusion

- There are various policies and act regarding environment protection in India which ensure conservation at various stages.
- EIA is an important for securing environment while development projects.
- EIA draft 2020, is upcoming EIA process regarding which there are some concerns among few experts.



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Thank You